

Case Assembly Guide

Required Parts

- ☐ Waveshare ESP32-S3-Touch-LCD-1.46C board
- ☐ 3D printed case (2 parts): [Flight Radar Enclosure.3mf](#) in the [/3mf](#) folder
- ☐ USB-C cable (for power/programming)
- ☐ (Optional) 3.7V LiPo battery for portable operation

Board Documentation

- **Official Waveshare Wiki:** <https://docs.waveshare.com/ESP32-S3-Touch-LCD-1.46>
- **Schematic PDF:** Available on Waveshare wiki under "Resources"
- **Dimensions:** ~49mm × ~49mm

Assembly Steps

Step 1: Test Board Before Assembly

Always flash and test the board BEFORE mounting in the case:

1. Upload FlightRadar.ino
2. Verify display shows "FlightRadar" splash
3. Test touch responsiveness
4. Connect to WiFi

Step 2: Case Orientation

- **Display faces up/out** (round LCD visible)
- **USB-C port** should align with case opening

Step 3: Mounting

- Use M2 x 4mm screws (3 locations)
- Do not overtighten

Step 4: Cable Management

- Route USB-C cable through enclosure back
- Battery (if used) fits in compartment behind board

Troubleshooting Assembly Issues

Issue	Solution
Screw holes don't line up	Check board orientation - board only fits one way
Back of case isn't staying on	Make sure to push it until it clicks into place

3D Print Settings

- Layer height: 0.2mm
- Infill: 15%
- Supports: Built into the design, so keep them turned off
- Material: PLA or PETG